

# A FIXED-POINT COUNTEREXAMPLE TO CONDITION-NUMBER-ONLY STABILITY OF SHERMAN–MORRISON ITERATIVE REFINEMENT

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**ABSTRACT.** Hashemi and Nakatsukasa conjectured that fixed-precision iterative refinement stabilizes the Sherman–Morrison solution of  $(A + uv^T)x = b$  when both  $A$  and  $A + uv^T$  have condition numbers safely below the reciprocal of unit roundoff. We disprove the natural uniform form of this condition-number-only conjecture. In precision- $p$  binary round-to-nearest arithmetic, an exact 2-by-2 family has  $\epsilon\kappa_2(A), \epsilon\kappa_2(A + uv^T) \rightarrow 0$ , yet every computed refinement iterate is the zero vector and has relative backward error exactly one. A binary64 member is certified using exact rational arithmetic and an independent hardware trace. We also derive the exact rank-one propagation factor caused by a reused rounded Sherman–Morrison denominator and a conditional positive recurrence. The counterexample does not contradict the source paper’s conditional one-step theorem or safeguarded variants of the algorithm.

## 1. INTRODUCTION

For  $B = A + uv^T$ , the Sherman–Morrison (SM) formula applies  $B^{-1}$  through solves with  $A$  and one scalar denominator. Hashemi and Nakatsukasa introduced a fixed-precision iterative-refinement variant, SM–IR, and proved backward stability under an explicit, verifiable one-step hypothesis. They further conjectured, based on experiments, that SM–IR eventually returns a backward-stable solution provided  $\kappa_2(A)$  and  $\kappa_2(B)$  are safely below  $\epsilon^{-1}$  [1].

The words “safely” and “eventually,” the stopping predicate, and the constant implicit in backward stability are not quantified in that prose conjecture. We therefore test its standard uniform meaning: as precision  $p$  varies, unit roundoff is  $\epsilon_p = 2^{-p}$ ; the hypotheses require  $\epsilon_p\kappa_2(A_p), \epsilon_p\kappa_2(B_p) \rightarrow 0$ ; and backward stability requires a bound  $C(n)\epsilon_p$  with  $C(n)$  independent of the data and precision. The counterexample has backward error one for every  $p$ , so it also violates any conventional fixed binary64 threshold, including the  $5\epsilon$  threshold used in the source experiments.

The obstruction is a fixed point, not overflow, underflow, division by zero, or a falsely accepted stopping test. The computed residual remains nonzero, but the reused SM corrector maps it to zero on every pass.

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## 2. ARITHMETIC AND ALGORITHM

Let  $\mathbb{F}_p$  denote a radix-two format of precision  $p \geq 2$ , with round-to-nearest, ties-to-even, and enough exponent range to contain all quantities displayed below. Its unit roundoff is  $\epsilon_p = 2^{-p}$ .

We use the operation graph of Algorithms 1–2 in the source paper. Given  $A, u, v, b$ , the initial SM solve computes

$$\begin{aligned}\hat{y} &= \text{fl}(A \setminus b), & \hat{z} &= \text{fl}(A \setminus u), \\ \hat{\alpha} &= \text{fl}(v^T \hat{y}), & \hat{\beta} &= \text{fl}(1 + \text{fl}(v^T \hat{z})), \\ \hat{\theta} &= \text{fl}(\hat{\alpha}/\hat{\beta}), & \hat{x}_0 &= \text{fl}(\hat{y} - \text{fl}(\hat{\theta} \hat{z})).\end{aligned}$$

At refinement step  $k$ , it forms the split residual

$$\hat{r}_k = \text{fl}(b - A\hat{x}_k - (v^T \hat{x}_k)u),$$

whose residual-error analysis fixes the evaluation graph

$$\hat{p}_k = \text{fl}(b - \text{fl}(A\hat{x}_k)), \quad \hat{q}_k = \text{fl}(\text{fl}(v^T \hat{x}_k)u), \quad \hat{r}_k = \text{fl}(\hat{p}_k - \hat{q}_k).$$

It then solves  $Ay_{r,k} = \hat{r}_k$ , and reuses the initial  $\hat{z}, \hat{\beta}$  to obtain the correction

$$\hat{d}_k = \text{fl}\left(\hat{y}_{r,k} - \text{fl}\left(\frac{\text{fl}(v^T \hat{y}_{r,k})}{\hat{\beta}} \hat{z}\right)\right).$$

The source Step 5 writes  $\hat{x}_k + \hat{y}_{r,k} - \hat{\theta}_{r,k} \hat{z}$  without fixing its floating-point parentheses. The verifier uses the literal left association  $\text{fl}(\text{fl}(\hat{x}_k + \hat{y}_{r,k}) - \text{fl}(\hat{\theta}_{r,k} \hat{z}))$  and independently checks the correction-first association  $\text{fl}(\hat{x}_k + \text{fl}(\hat{y}_{r,k} - \text{fl}(\hat{\theta}_{r,k} \hat{z})))$ . Both are exactly zero on the certificate; the result is also unchanged by ordinary fused evaluation.

For a finite approximate solution  $x$ , define the normwise relative backward error

$$(1) \quad \eta_B(x) = \frac{\|b - Bx\|_2}{\|B\|_2\|x\|_2 + \|b\|_2}.$$

At the zero vector with  $b \neq 0$ , this quantity is exactly one. If only matrix perturbations are allowed, zero is even less defensible: no  $\Delta B$  can make  $(B + \Delta B)0 = b$ .

## 3. EXACT FIXED-POINT COUNTEREXAMPLES

**Theorem 3.1** (Scalar family). *For every admissible precision  $p$ , take*

$$A = [1], \quad u = [1], \quad v = [2^p], \quad b = [1].$$

*Then  $B = A + uv^T = [2^p + 1]$ ,  $\kappa_2(A) = \kappa_2(B) = 1$ , and the literal SM-IR iteration satisfies*

$$\hat{x}_k = 0, \quad \eta_B(\hat{x}_k) = 1 \quad \text{for every } k \geq 0.$$

*Consequently the method is not uniformly backward stable under the two condition-number hypotheses alone.*

*Proof.* At the start of the binade containing  $2^p$ , adjacent precision- $p$  numbers are  $2^p$  and  $2^p + 2$ . Their midpoint is  $2^p + 1$ . The lower candidate has even integer significand  $2^{p-1}$ , whereas the upper candidate has odd integer significand  $2^{p-1} + 1$ . Therefore

$$(2) \quad \text{RN}_{\text{even}}(2^p + 1) = 2^p.$$

Every other operation in the initial trace is exact:

$$\hat{y} = \hat{z} = 1, \quad \hat{\alpha} = 2^p, \quad \hat{\beta} = 2^p, \quad \hat{\theta} = 1, \quad \hat{x}_0 = 1 - 1 = 0.$$

If  $\hat{x}_k = 0$ , the split residual is exactly  $\hat{r}_k = 1$ . The correction repeats the same trace:  $\hat{y}_{r,k} = 1$ ,  $\text{fl}(v^T \hat{y}_{r,k}) = 2^p$ , and the correction is  $1 - 1 = 0$ . Thus zero maps to zero, and induction proves the assertion for all iterations. Equation (1) gives  $\eta_B(0) = 1$ . Finally,  $\eta_B(0)/\epsilon_p = 2^p \rightarrow \infty$ , ruling out any precision-independent backward-stability constant.  $\square$

The scalar family already meets the conjectured hypotheses with the largest condition-number margin. The next embedding shows that the mechanism is not an artifact of scalar condition numbers being identically one.

**Theorem 3.2** (Two-dimensional safely-conditioned family). *Let  $p \geq 4$ ,  $m = \lfloor p/2 \rfloor$ , and*

$$A_p = \text{diag}(1, 2^m), \quad u_p = b_p = e_1, \quad v_p = 2^p e_1.$$

*Then*

$$B_p = \text{diag}(2^p + 1, 2^m), \\ \kappa_2(A_p) = 2^m, \quad \kappa_2(B_p) = 2^{p-m} + 2^{-m},$$

*and hence*

$$\epsilon_p \kappa_2(A_p) \rightarrow 0, \quad \epsilon_p \kappa_2(B_p) \rightarrow 0.$$

*With correctly rounded diagonal solves, every SM-IR iterate is zero and has backward error one.*

*Proof.* The singular values and condition numbers follow directly from the positive diagonal entries. The solves with  $A_p$ , all dot products, and all updates remain in the first coordinate and are exact. Thus the computation is the trace of Theorem 3.1 embedded in the invariant first coordinate. The limit statements follow from

$$\epsilon_p \kappa_2(A_p) = 2^{m-p}, \quad \epsilon_p \kappa_2(B_p) = 2^{-m} + 2^{-p-m}.$$

$\square$

For binary64,  $p = 53$  and  $m = 26$ . The exact certificate is

$$A = \text{diag}(1, 67108864), \quad B = \text{diag}(9007199254740993, 67108864),$$

with

$$\kappa_2(A) = 2^{26}, \quad \kappa_2(B) = 2^{27} + 2^{-26},$$

whereas  $\epsilon^{-1} = 2^{53}$ . All nonzero computed values are normal and finite;  $\hat{\beta} = 2^{53} \neq 0$ .

#### 4. STRUCTURAL SOURCE OF THE FIXED POINT

The next two lemmas separate what the condition numbers permit from what the rounded corrector does.

**Lemma 4.1** (Denominator range). *Let  $n \geq 2$ ,  $B = A + uv^T$ , and  $\beta = 1 + v^T A^{-1} u \neq 0$ . Then*

$$\frac{1}{\kappa_2(A)\kappa_2(B)} \leq |\beta| \leq \kappa_2(A)\kappa_2(B).$$

*The family in Theorem 3.2 attains the upper bound.*

*Proof.* Choose nonzero  $q \in \ker v^T$ . Since  $Aq = Bq$ , comparison with the smallest singular values gives

$$\frac{\|B\|_2}{\|A\|_2} \leq \kappa_2(B), \quad \frac{\|A\|_2}{\|B\|_2} \leq \kappa_2(A).$$

The rank-one matrix  $BA^{-1} = I + uv^T A^{-1}$  has exceptional eigenvalue  $\beta$ . Hence

$$|\beta| \leq \|BA^{-1}\|_2 \leq \kappa_2(A) \frac{\|B\|_2}{\|A\|_2} \leq \kappa_2(A)\kappa_2(B).$$

Apply the same argument to  $AB^{-1}$ , whose exceptional eigenvalue is  $\beta^{-1}$ , for the lower bound. In Theorem 3.2,  $\beta = 2^p + 1 = \kappa_2(A_p)\kappa_2(B_p)$ .  $\square$

Thus separate assumptions  $\epsilon\kappa_2(A), \epsilon\kappa_2(B) \ll 1$  still allow  $\epsilon|\beta|$  of order one. A product condition can block this specific mechanism, but we do not claim it is sufficient for the fully rounded algorithm.

**Lemma 4.2** (Reused-denominator propagation). *Assume  $A$ -solves and vector operations are exact, but the corrector reuses a fixed nonzero  $\widehat{\beta}$  in place of*

$$z = A^{-1}u, \quad t = v^T z, \quad \beta = 1 + t.$$

*Put  $d = \widehat{\beta} - \beta$ . The corrector*

$$M = A^{-1} - \frac{zv^T A^{-1}}{\widehat{\beta}}$$

*has residual-propagation matrix*

$$R = I - BM = -\frac{d}{\widehat{\beta}} uv^T A^{-1}.$$

*It satisfies  $R^2 = \lambda R$ , where*

$$\lambda = -\frac{dt}{\widehat{\beta}}.$$

*Consequently, except on the nullspace of  $R$ , exact-residual refinement contracts precisely when  $|\lambda| < 1$ .*

*Proof.* Direct multiplication gives

$$BM = I + uv^T A^{-1} - \frac{(Bz)v^T A^{-1}}{\widehat{\beta}} = I + \frac{d}{\widehat{\beta}} uv^T A^{-1}.$$

The rank-one identity follows by one more multiplication, using  $v^T A^{-1}u = t$ . If  $r_{k+1} = Rr_k$ , then  $r_{k+1} = \lambda^k r_1$  after the first step.  $\square$

In the counterexample,  $t = 2^p$ ,  $\beta = 2^p + 1$ ,  $\widehat{\beta} = 2^p$ , and  $d = -1$ , so  $\lambda = 1$  and  $M = 0$ . Notice that neither denominator is small; a nonzero-denominator safeguard alone does not repair the conjecture.

## 5. A CONDITIONAL POSITIVE RECURRENCE

Although the two condition numbers do not imply contraction, the recurrence suggested in the conjecture follows once contraction is independently verified.

**Proposition 5.1.** *Let  $r_k = b - Bx_k$  be the exact residual. Write one implemented step as*

$$\hat{r}_k = r_k + f_k, \quad c_k = \hat{r}_k - Bd_k, \quad x_{k+1} = x_k + d_k + s_k.$$

If

$$\|c_k\|_2 \leq \rho \|\hat{r}_k\|_2, \quad 0 \leq \rho < 1,$$

and, for a fixed scale  $D > 0$ ,

$$\|f_k\|_2 \leq C_r \epsilon D, \quad \|B\|_2 \|s_k\|_2 \leq C_a \epsilon D,$$

then  $\zeta_k = \|r_k\|_2/D$  satisfies

$$\zeta_{k+1} \leq \rho \zeta_k + C\epsilon, \quad C = (1 + \rho)C_r + C_a,$$

and therefore

$$\zeta_k \leq \rho^k \zeta_0 + \frac{C}{1 - \rho} \epsilon.$$

*Proof.* Substitution gives the exact identity

$$r_{k+1} = -f_k + c_k - Bs_k.$$

The stated one-step inequality follows from the triangle inequality and

$$\|\hat{r}_k\|_2 \leq \|r_k\|_2 + \|f_k\|_2.$$

Summing the geometric recurrence proves the final bound.  $\square$

The missing premise is exactly the uniform contraction  $\rho < 1$ . The counterexample has  $\rho = 1$ ; condition numbers alone cannot imply the premise. A full fixed-precision theorem must additionally control residual representation growth and update rounding relative to the selected backward-error denominator.

## 6. CERTIFICATION, LIMITATIONS, AND REPRODUCIBILITY

The serialized binary64 certificates contain only integers and rationals. The primary verifier reconstructs round-to-nearest with ties to even using Python integer and rational arithmetic and proves the midpoint decision, exact condition numbers, the transition  $T(0) = 0$ , and backward error one. A separate C program checks the scalar trace on IEEE binary64 hardware. The logical proof of perpetual stagnation is the one-step fixed-point identity plus induction; finite iteration counts in the scripts are redundant sanity checks.

The verifiers parse certificates fail-closed. Every top-level and nested object has an exact key set, all values have exact JSON types, duplicate keys and nonstandard numeric constants are rejected, and the input bytes are bound to an explicit SHA-256 trust anchor. An adversarial suite checks that incorrect hashes, extra or dropped keys, and changes to the input, trace, or arithmetic declaration all exit nonzero. Semantic mutants are supplied with their own correct hashes, so these tests reach the schema and exact recomputation checks rather than stopping only at the hash gate.

The result concerns the literal split SM-IR operation graph and reuse of  $\hat{z}, \hat{\beta}$ . It does not contradict the conditional one-step theorem in the source paper. It also

does not address algorithms that rescale or refactor the update, form  $B$ , recompute the corrector, use compensated or higher-precision arithmetic, or fall back to a stable direct solve. Any repaired positive theorem must control the full correction residual, not merely the sizes of the two spectral condition numbers or a lower bound on  $|\beta|$ .

Accordingly, the theorem refutes only the original broad conjecture under its natural uniform condition-number-only reading. A mere  $\hat{\beta} \neq 0$  no-breakdown test does not exclude the example, and ordinary fused operations do not change it. A repaired conjecture that adds a scale-sensitive lost-addend/cancellation guard, verifies contraction, uses extra precision, or invokes a stable fallback is a different statement and remains open.

All artifacts are in the accompanying project. The principal checks are

```
python3 verification/verify_stagnation.py \
  certificates/binary64_stagnation.json
python3 verification/verify_stagnation_2x2.py \
  certificates/binary64_stagnation_2x2.json
python3 verification/test_tamper_fail_closed.py
cc -std=c11 -O0 -frounding-math -ffp-contract=off \
  verification/verify_stagnation_hardware.c -lm \
  -o verification/verify_stagnation_hardware
./verification/verify_stagnation_hardware \
  certificates/binary64_stagnation.json
```

No proof assistant was used. Generative-AI agents assisted with proof and counterexample search, statement repair, exact-verifier implementation, literature search, and adversarial review. Every decisive claim in the paper is an exact arithmetic identity, an induction, or is reconstructed by the disclosed verifiers; numerical search is not used as proof.

## REFERENCES

1. Behnam Hashemi and Yuji Nakatsukasa, *Instability of the sherman-morrison formula and stabilization by iterative refinement*, 2025.

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